Carburettor?

- The first carburettor was invented by Samuel Morey in 1826.
- A carburettor meters out and mixes fuel with air for a proper combustible mixture.
- carburettors last longer than fuel injection systems and are favoured in motor sports.



9.3.1 Fuel Petrol Pump

The work of the petrol pump is to suck the petrol from the petrol tank with pressure and send it to Carburettor. Generally by the way of working it can be classified in to the types. They are,

1. AC mechanical petrol pump
2. SU Electrical petrol pump

9.3.1.1 AC mechanical petrol pump

The power required to drive the pump is taken from the eccentric in the cam shaft. So it is placed near cam shaft in cylinder block. When the engine starts the crank shaft starts to rotate. The cam shaft is actuated by the timing gear connected to it. The eccentric in the cam shaft operates the rocker arm in the pump. As the rocker arm moves upwards the diaphragm to it pushes the pull rod downwards. At that time the Vacuum is created in the pump chamber and the inlet value is being opened as a result. By which the petrol is being sucked in at this time the outlet valve is at clocked stage.

As the cam shaft further rotates the rocker arm is released from the eccentric pressure. So that with the help of the spring the pull rod moves the diaphragm to older stage. As a result the petrol in the pump chamber pressed out of the outlet valve and reaches floating chamber.

When the float chamber in the carburettor is filled with petrol, the needle valve closes the inlet path. So that the petrol can't come out of the outlet valve. So that the pressure is raised inside the pump chamber and the pressure makes the diaphragm and pull rod to stay down itself. At that time even the pull rod won't be actuated as a result the petrol won't come out of petrol chamber.

By this condition, the engine continuous to rotate the petrol is being supplied so that the petrol in the float chamber gets reduced and the needle valve is being opened, which makes the fuel to come out from the tank. So that the pressure in the pump gets reduced and starts to work as earlier. So that all operating conditions of engine the pump works smoothly

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